

Development of Airbag Fabrication Machine and Process for Physical Human Machine Interaction

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Abstract—In this paper, we present an airbag fabrication machine and its process to make airbags which can be utilized for human-machine interaction. The proposed machine uses a CNC-style gantry with XYZ axes to seal layers of thermoplastic sheets in intricate patterns. A custom designed ball transfer is mounted on this machine to control temperature to press, heat, and fuse the thermoplastic sheets. We also implemented a path planning algorithm for heat sealing based on user's drawing. Using the developed machine and process, we demonstrate fabricated airbags can be pressed more than our target pressure and durable to be used in multiple applications over time.

I. INTRODUCTION

Utilization of pneumatic structures to protect humans or objects, such as equipping a vehicle with airbags and locating air-filled cushions in a package, is very common in everyday life. Also, one can make pneumatic structures to be actuated with a compressor. However, fabrication of a simple pneumatic structure is sometimes not an easy task, especially when its shape and operating pressure are given for a specific usage. In this paper, we present an airbag fabrication machine and its automated process to make airbags, which can be used for physical human machine interaction.

In the robotics field, the demand for physical interaction between human and robot has been growing in industry, entertainment, education, therapy and so on. For safe human-robot interaction, a robot should be compliant and reactive to avoid damaging its hardware and human injury. Recently, there are interesting efforts to test novel mechanism utilizing soft materials to make a robotic system compliant and soft passively and actively [1]–[3]. We also previously developed 3D printed soft skin modules and heat-sealed airbags which sense force via an air-filled cavity connected to a pressure sensor [4]–[7]. The force sensing modules developed are designed to cover the various links of a robotic system, enabling a robot to sense force over a large area of its body while requiring less cumbersome electronics and wiring than other tactile sensor networks. In these studies, both 3D printed modules and pressurized heat-sealed polyurethane modules are effective in physical interaction. However, there are still many difficulties in the fabrication process and material selection which can be improved.

Heat sealing can be done by hand using two main tools, a heated hand roller or an impulse sealer. The resulting seals are limited to straight lines, with slight curvature possible

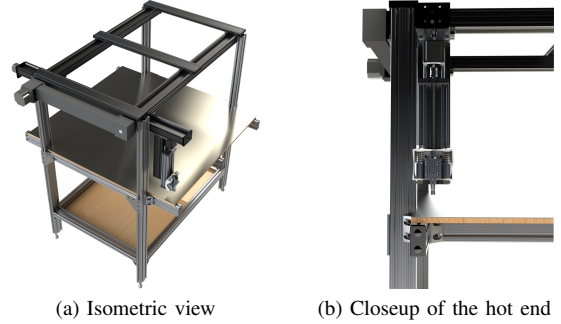


Fig. 1. Overview of heat Seal table and hot end



Fig. 2. Sealed and inflated airbag

with a heated hand roller. This imposes a serious limitation: complex shapes cannot be made with hand methods. There are previous efforts utilizing CNC (Computer Numeric Control) style machines to make complex shapes accurately [8]–[11]. Creating complex shapes allows for controlling the behavior of an inflated bladder. Niiyama et al. [8]–[10] designed heat sealing machines to fabricate air pouches in order to utilize them as actuators in robotic systems. Aeromorph [11] was implemented to investigate how to make origami structures with inflatables with paper, plastics and fabric. Inspired by these previous studies, we designed our machine to seal complex shapes in polyurethane, resulting in an air bladder (airbag). We focus on finding a proper design to make airbags for human-machine interaction, aiming for the following goals:

- 1) Design process should be user friendly and automated.
- 2) Airbags should be large enough to attached to a contoured surface, such as a human arm or chest.
- 3) Airbags should remain unchanged up to certain pressure and over time.

This paper is organized as follows. The physical design of the heat sealing machine is presented in Section II. In Section III, we describe the path planning algorithm,



Fig. 3. Heat sealing table assembly

software interface, and workflow. Section IV and V present experimental implementations, results of airbag tests and application examples. Finally, conclusions and future work are discussed in Section VI.

II. HEAT SEALING MACHINE DESIGN

Our approach for heat sealing designs was based on designing and building a CNC machine with a large work area (Fig. 1).

A. Gantry Table Design and Assembly

The moving part design was based on an existing X-Y Gantry commercially available from Velmex, Inc [12]. This is supported above a table top with an 80/20 aluminum extrusion support structure. The Z-axis, which holds the heat sealing element, uses a commercially available motorized linear axis [13], which is bolted to a post connected perpendicular to the X-Y gantry. The three axes are controlled by two Velmex VXM motor controllers, allowing for two of the motors to be controlled simultaneously. The tabletop is a laser cut piece of 3/4" birch plywood. A 16 gauge steel sheet was attached to the plywood to allow the use of magnets on the table and to provide a surface suitable for heat sealing. The heat sealing table has an approximate work area of 90 cm by 90 cm, which provides enough space to fabricate air bags of our desired size.

Assembling the table was broken down into three steps. First, the plywood table top was adhered to the steel sheet using double sided carpet tape. The table top was flipped upside down to attach the 80/20 aluminum extrusions that support it. The four main posts were then slid into place. The bottom support bracket was slid onto the main posts and secured at about 200 mm from the top. The assembly was then flipped right side up and the table secured into place, about 700 mm from the top. Lastly the X-Y-Z gantry was placed on top of the table assembly, completing the machine assembly in Fig. 3.

B. Hot End Design

We needed the element that physically heat seals and makes contact with the material to have little resistance as

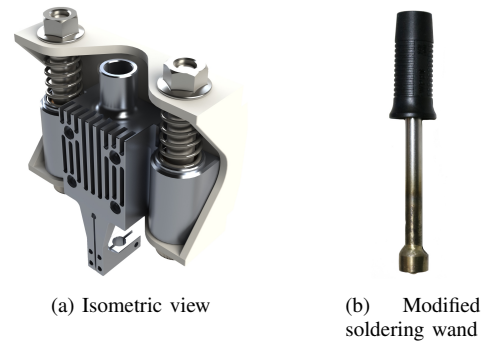


Fig. 4. Hot end overview

it moves atop the work piece. The element should also be able to seal small designs, so that the precision of the X-Y gantry (0.001 inches) [12] could be utilized. The thickness of the heat seal directly influences the detail that can be sealed. Since we want to create fine details, the heat sealing element should have a relatively small size. We chose a ball transfer, from Misumi USA [14], as the physical heat sealing element. Ball transfers have a large ball bearing surrounded by several smaller, loose bearings. This allows for low friction as the large ball bearing moves. The ball bearing imparts both heat and pressure to the workpiece, creating a heat seal.

To heat the ball transfer, we used a temperature controlled soldering station. The soldering station outputs up to 85 watts of heat to a wand connected to the base. The wand has a removable steel sleeve, allowing a user to change the soldering tips. We made a modified soldering wand, responsible for sealing the plastic layers, by combining the soldering wand and the ball transfer (Fig. 4b). The stem on the ball transfer was shortened, the stem inserted into the wand sleeve, and then the ball transfer and the stem were brazed together. The sleeve can then be screwed onto the wand, where the heating element of the wand contacts and transfers heat to the ball transfer.

Figure 4a shows a 3D-printed mechanism, designed to hold the modified wand and connect it to the CNC Z-axis. As the ball transfer rolls across the surface of the work piece while sealing, it may pass over ripples in the material or already existing seals. To minimize the effect of these possible disturbances, a compliant mechanism was used. Two precision ground screws act as rails allowing the wand holder to slide, a spring acts on the wand sleeve holder to push it down. The wand holder has is made up of two 3D-printed parts which are contoured to the soldering wand handle and are bolted together. When the Z-axis moves down to begin the heat seal, the springs allow for some vertical movement if the material has uneven surface or ripples. This also adds another safety benefit that if there is an error with the machine, like a limit switch malfunction, the spring can take up the extra movement without damaging the Z-axis gantry. The mechanism also has tines that grip the wand close to the ball transfer, preventing the sleeve stem from moving under the X-Y gantry's movement. These design considerations

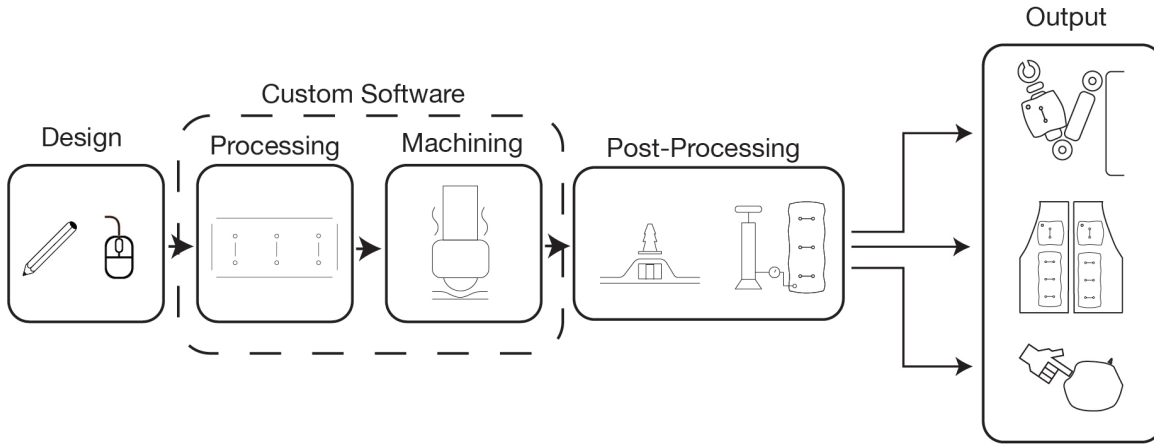


Fig. 5. Overview of design and fabrication process

were to keep the ball transfer as rigid as possible. With earlier designs, the lack of rigidity caused the wand stem to move in relation to the Z-axis. That is, instead of staying parallel to the Z-axis and thus perpendicular to the tabletop surface, the stem was about 10 degrees off, opposite of the direction of travel. This caused the heat sealed design dimensions to not match up with original dimensions.

III. SOFTWARE

One of our goals is the realization of user friendly and automated design process. For this goal, we developed custom software, shown in the process in Fig. 5, to have an easy user interface and to automate heat sealing. To control the CNC gantry, we created a GUI (Graphical User Interface) into which one can insert their design file, see a preview of the design, and a can start a heat seal job. The GUI also contains buttons for a maintenance position and to calibrate the Z-axis. The computer software controlling the X-Y-Z gantry is a MATLAB script with an accompanying GUI. It uses built in serial libraries to communicate with the CNC motor controllers.

A. Workflow

The workflow starts with a user's design. The design is made in a vector graphics or CAD drawing application in order to make it simple to modify dimensions. The design should be black and white as the contrast allows the program to accurately convert the design to coordinates. When designing the pattern, attention should be placed onto centering the design in the page. The page size should reflect the size of the thermoplastic sheet to be used, so the design should include a buffer zone of at least 1.5 inches between the page boundary and the design. This design is exported into a PDF file, then rasterized at 100 pixels per inch. The specific resolution correlates to the resolution desired from the X-Y-Z gantry. The rasterized image, such as a .png file, is imported by the script, which converts the image into coordinates using a MATLAB function called `pic2points` [15]. The coordinates are in pixels and are converted to inches using the pre-defined resolution (in pixels per inch).

The coordinate information is then ordered to make "paths" or a list of coordinates that will create a heat sealed line. Each path is then converted in to commands that the motor controllers can understand and finally sent to the controller.

B. Creating Paths

Recognizing and creating paths is an integral part of the software. To accomplish this, a heuristic algorithm was created (Alg. 1).

The algorithm is structured so that the result would be similar to how a human would make a desired shape. For example, a test shape used in creating this algorithm can be seen in Fig. 6a, where the colored lines represent the individual paths to make the design. Intuitively, the outside would be drawn first, then each individual internal line. This is important because of the way heat sealing works; the hot end is rolled across the surface in the same way as a ballpoint pen. Thus, shapes must be sealed in continuous paths and not discretely, like in an ink jet printer.

Algorithm 1 Heuristic algorithm to sort points

- 1: initialize leftover list and path list
 - 2: fill leftover list with all points
 - 3: start at top left pixel
 - 4: **while** point exists in leftovers **do**
 - 5: create new path list
 - 6: **while** distance to any point < threshold **do**
 - 7: find closest point
 - 8: remove point from leftovers
 - 9: place point in current path list
 - 10: **end while**
 - 11: **end while**
-

Before the algorithm begins, a leftover list and a paths list must be created. The leftovers list contains all unsorted coordinates, while the paths list contains ordered coordinates for a single path. A design may range from a single path, for a simple geometric object, to tens of paths for complex designs. The algorithm follows a path within the shape from an initial point. The initial point is the top leftmost point in

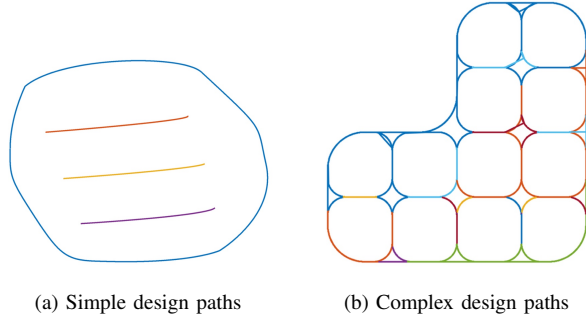


Fig. 6. Design examples

the image, due to how the coordinate values are arranged by `pic2points` [15]. This point becomes the first entry in the first path list. From the initial point, the next closest point is found and appended to the paths list. This becomes the new current point. Finding next closest point continues until there are no more close points to the current point. At the end of a path, if the start and points are close, denoting a closed shape, then the start point of that path is appended to the list and becomes the new last point. This “closes” the otherwise open shape, improving the heat seal quality. If there are still coordinates left in the leftover list, the top leftmost point is chosen as a new start point and a new path list is created. The path finding process continues until there are no more points in the leftover list.

Applying the algorithm on the simple heat seal design example in Fig. 6a, the first path found is the outer circle. The second is the top horizontal line, then the middle line, and last the bottom line. The algorithm works well with simple shapes. Complex shapes, however, make many more disjointed paths because they contain a large number of shared points, as shown in Fig. 6b. If a line splits into two or more lines, the algorithm will end up choosing the closest one, without knowing if it will be the best path. This is clear in Fig. 6b where each different color is a different path the algorithm chose to recreate the design. There are many disconnected paths, in contrary to how a human might recreate the design; by grouping the internal shapes together as one path and the outside border as another. Thus our algorithm resembles a greedy algorithm; it blindly follows the closest path without knowing if it would result in a heat seal with less paths.

Implementing this algorithm creates an organized collection of the paths which, when run in order, enable the heat sealing tip to recreate the design. The threshold to consider a point “close” was determined to be 0.05 inches experimentally and from the predetermined resolution of the design. The paths are then converted into commands by adding motor speed information and formatting the coordinates in each path. The coordinates are scaled to match the steps per inch that the gantry moves at. Commands are then arranged in the order they in which they were created by the algorithm. All the commands are saved to a plain text file with a

`.htsl` extension. This enables repeatability, reducing the time needed to heat seal multiple designs. Communicating with the VXM controllers, which control the stepper motors on the gantry, is done by sending one line commands and waiting for a completion response, “^”, from the controller. Commands are sent one at a time because the controller does not have enough memory to store every command for a complete design. The default serial communication libraries in MATLAB were used to send commands and read responses to the motor controllers through a USB RS-232 interface.

IV. HEAT SEALING AIRBAGS

A. Material

Airbags are made from a thermoplastic polyurethane (TPU) plastic sheet purchased in bulk from Polyurethane Products [16]. The 15 mil (0.015 inch) material proved to be robust due to its thickness and the properties of the plastic. We tried heat sealing thinner TPU (8 mil), HDPE, and PVC, however only 15 mil TPU and PVC heat sealed reliably. This is in contrast to previous work on heat sealed inflatables [11], where they were able to seal thinner and a wider variety of materials.

TPU sheets are prepped for heat sealing by cutting an appropriate sized piece from the material stock. The sheet sizing comes from the page size of the design, or can be approximated from the GUI preview. The TPU sheets were cut on a cutting mat with an inch grid, allowing us to quickly prepare a designs material. Because of the manufacturing process for the TPU sheets, the material has two surface finishes: a matte and a glossy side. For more consistent seals, we placed the two glossy sides together before sealing.

B. Temperature

The temperature we used to seal 15 mil TPU was 450 °C, the maximum setting of the soldering station. While the heating element reaches 450 °C, the temperature at the ball transfer is actually about 230 °C. The large heat loss was from the surface area of the ball transfer; the soldering station was designed for soldering tips, which have a much smaller surface area. The melting temperature for TPU is between 106 and 227 °C [17], meaning the ball transfer does indeed get hot enough to melt and seal the plastic film. The speed of CNC gantry, about 152.4 cm/min, was just slow enough to fully melt and fuse the two plastic layers together as the ball transfer passed over the work piece.

C. Modified Barbed Tube Fitting

To ensure that the airbags can be pressurized while remaining air-tight, a modified barbed fitting for a 1/16” ID tube was used. The modified fitting consists of a thread-forming nylon nut, O-ring, and flanged barbed fitting. For the TPU sheet to be compressed effectively between the nylon backing nut and the fitting, the fitting needed to be flanged. Since there are no commercially available solutions at that small scale, we joined a brass washer to the barb by soldering the two together. Soldering the fitting and washer together formed



Fig. 7. Airbag fitting

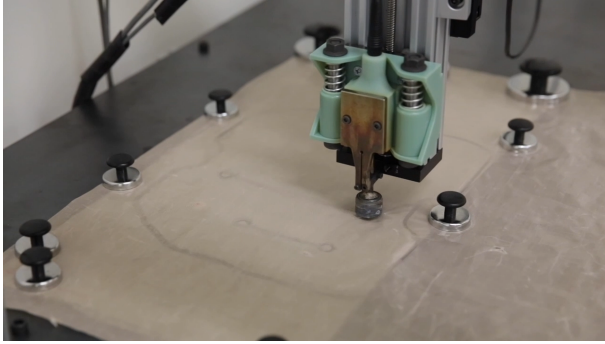


Fig. 8. Machine mid heat seal

an air tight seal that was both permanent and strong. An alternative to soldering the two is super glue, like the barb in Fig. 7, which we used in earlier airbags. However, super glue is sensitive to temperature and can form cracks when it dries, resulting in a barb that was not air tight. Super glue also dried on the threads of the fitting, preventing the fitting from screwing into the nylon nut. For a better seal, we used an x-profile O-ring between the flanged fitting and the nut.

D. Heat Sealing Process

Prior to heat sealing, the nylon backing nut must be placed between the TPU sheets. The location for the nut was approximated from the software GUI by selecting a blank space away from heat seal lines and marking it with a marker. This was done on a cutting mat with an inch grid, ensuring the locations was marked correctly. The nut displaces the TPU material when placed because it lifts the two sheets apart. To minimize this, we put the flanged side of the nut down, reducing the horizontal displacement. A PTFE-coated fiberglass sheet [18] was placed over the TPU layers to ensure the polyurethane did not stick to the heated ball-transfer.

Securing the prepared sheet to the heat seal table was accomplished with various magnets (Fig. 8). Because of the Z gantry's height from the table, care must be taken when placing the larger magnets as to prevent them from hitting the gantry. Thus, the bigger magnets are placed on the top left and both right corners. Two or more smaller magnets are used to hold the bottom left hand corner in place. Before placing the magnets, the TPU and fiberglass layers are smoothed down and pulled tight to prevent ripples. Once the piece has been secured, the GUI is used to start the heat sealing

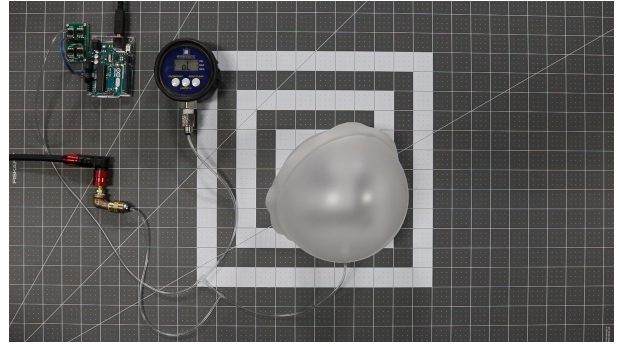


Fig. 9. Experimental setup for airbags

hot end moving along the paths created from the specified design. A typical heat seal job lasts about 5 minutes, with the time increasing to 10 or more minutes for large and complex designs.

E. Airbag Assembly

Finishing up a airbag involves screwing in the modified barbed fitting into the nylon nut. The nut is sealed within the airbag; to access the nut we used a screwdriver style hex driver and, in a screwing motion, pushed down on the fitting centered on the nut. The threads on the barbed fitting cut a hole in the material. The barb was then unscrewed, the O-ring was placed on the barb, and the barb was screwed down again. Care should be taken when screwing down onto the x-profile O-ring as it had a tendency to deform and spill out from under the barb's flange. Once that is done, the airbag is then ready for air to be pumped into it.

Air was pumped into airbags using a commercial bike pump that had a one way check valve. This allowed us to inflate airbags without the air escaping since our barbed fittings did not have check valves. If the barbed fitting was not properly installed, air would leak right away. For most tests, we filled the airbags to 2 psi (13.79kpa) of pressure, checking the pressure with an external pressure gauge. Once inflated, airbags could then be connected to an analog pressure sensor or capped to preserve the air pressure.

V. TESTING AIRBAGS

Informally, sealed airbags, once inflated, were left untouched for over 24 hours. After the time elapsed, the airbags were physically inspected and the pressure measured. From these informal tests, we determined that the weakest point of the airbag is the seal between the modified fitting and the airbag material. Because the quality of the seal could not be visually determined, we used leak detection liquid [18], a liquid that airbags if a leak is present, to check fitting seal for leaks.

A. Setup

The destructive tests were recorded on a Honeywell pressure sensor (0-15 psig ; 0-103.42kpa) and visually using a 5 psig (34.47kpa) digital gauge and recorded with video camera, as shown in Fig. 9. The dual gauge setup was desired

TABLE I
MAX AIRBAG PRESSURES

	Trial	Max (PSIG)	Max (Kpa)
Circular airbags	1	6.299	43.435
	2_1	6.373	43.940
	2_2	5.823	40.149
	2_3	5.738	37.080
	3_1	6.318	43.561
	3_2	5.328	36.737
	3_3	5.250	36.197
	4_1	6.391	44.067
	4_2	4.412	30.418
Square airbags	1	4.738	32.667
	2	6.238	43.008
Mean Max Pressure		5.610	38.680
Standard Deviation		0.884	6.095

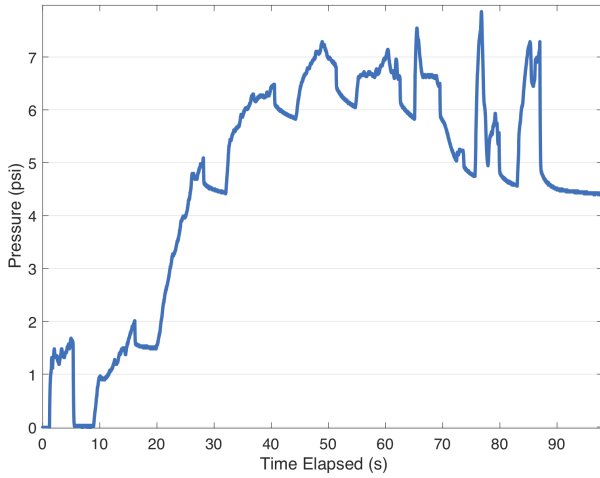


Fig. 10. Destructive test pressure graph sample

as to verify the pressure read from the Honeywell sensor. The pressure sensor was connected to an Arduino Uno, which sent the data through serial to a computer. The computer logged every serial message sent for the duration of the experiment. Airbags were slowly pumped with a hand pump as to not create large pressure spikes. Two different shapes of airbags were tested: a 15 cm diameter circle and a 15 cm by 15 cm square. Barbed fittings were primarily placed in the center of airbag for these trials. We first tested four circular airbags, then we tested two square airbags

B. Results

The second circular airbag's pressure during the destructive test can be seen in Fig. 10. The first characteristic of the graph is the pressure spikes from the pumped air. This is apparent through every test performed because as the handle is pushed down, the pump creates high pressure in the tubing connected to the pressure sensors. The second characteristic of the graph is downward trend of the graph as time increases. The elasticity of the TPU sheet causes the airbag to deform as the pressure increases. The elastic defor-



(a) Force jacket



(b) Robot skin

Fig. 11. Application examples

mation increases the internal volume of the airbag, causing the internal pressure to decrease. The material deformation also causes the cutout created for the fitting in the assembly process to widen. Even with the use of the x-profile O-ring, this leads to the fitting losing its seal and the airbag failing. Thus, the "burst pressure", or the final pressure of the airbag before failure, cannot really be taken as a comparable pressure for an airbag. Of the airbags tested, the only one that failed at the heat seal seam, and not at the fitting, was the first airbag.

To measure a pressure value that can be compared between airbags, maximum pressure experienced was recorded in Table I. This pressure is the maximum pressure recorded after any of the pressure spikes generated from pumping. Using this metric, we observed that the average airbag pressure before failure was 5.61 psig (38.68kpa). This average pressure was more than enough for our desired regular usage under 3 psig (20.68), found in [6].

C. Test in Applications

The fabricated airbags have been used long-term applications in our lab.

In [19], a force jacket in Fig. 11a was designed for a haptic wearable research. Throughout the jacket, twenty-six airbags were installed: six air compartments on the front of the jacket, six on the back, two on either side of the jacket, and four on each arm. A compressor supplies air to each airbag through individually controlled solenoid valves. And a 1.5"x1.5" square force sensitive resistor (FSR) was

placed in a pocket on each of the airbags on the inside of the jacket facing the body surface. The airbags were fully functional after intense use for the research, which includes over a month of 17 subject studies. This is a good example which shows the developed method in this paper is able to fabricate durable airbags for human machine interaction.

Another example of application is robot skin shown in Fig. 11b. In our previous work [7], a robotic arm system with airbags had been presented. The airbag fabrication process in the work was complicated and hard to duplicable. Using the machine and process in this paper, we could make robot skin easily and repeatably. To make a cylindrical shape, the airbags were designed with a pattern on it. For the forearm, velcro was stitched to one patterned airbag to cover the robot. Upper arm skin has two airbags stitched together. We designed simple 3D printed brackets to keep the airbags located at the mounted position. More detail can be found in the submitted video. This robot skin has been used over a year to run human-robot interaction studies. Filling up some air is needed about once a month to maintain the pressure level but the airbags have not broken down.

VI. CONCLUSION

This paper introduced an airbag fabrication machine and its process to make airbags with details. The goals were set in order to easily make durable airbags for human-machine interaction. We designed and implemented a machine uses a CNC-style gantry with XYZ axes to seal layers of thermoplastic sheets in intricate patterns. A custom designed ball transfer and barbed fitting were used to meet the goals. We also implemented a path planning algorithm for heat sealing based on users drawing. Using the developed machine and process, we demonstrated fabricated airbags are durable enough to be used in human machine interaction.

REFERENCES

- [1] R. Weitschat, J. Vogel, S. Lantermann, and H. Hoppner, "End-effector airbags to accelerate human-robot collaboration," in *2017 IEEE International Conference on Robotics and Automation (ICRA)*, May 2017, pp. 2279–2284.
- [2] A. Stilli, L. Grattarola, H. Feldmann, H. A. Wurdemann, and K. Althoefer, "Variable stiffness link (vsl): Toward inherently safe robotic manipulators," in *2017 IEEE International Conference on Robotics and Automation (ICRA)*, May 2017, pp. 4971–4976.
- [3] T. Kim, S. J. Yoon, and Y. Park, "Soft inflatable sensing modules for safe and interactive robots," *IEEE Robotics and Automation Letters*, vol. 3, no. 4, pp. 3216–3223, Oct 2018.
- [4] J. Kim, A. Alspach, and K. Yamane, "3d printed soft skin for safe human-robot interaction," in *Intelligent Robots and Systems (IROS), 2015 IEEE/RSJ International Conference on*, Sept 2015, pp. 2419–2425.
- [5] A. Alspach, J. Kim, and K. Yamane, "Design of a soft upper body robot for physical human-robot interaction," in *Humanoid Robots (Humanoids), 2015 IEEE-RAS 15th International Conference on*, Nov 2015, pp. 290–296.
- [6] J. Kim, A. Alspach, I. Leite, and K. Yamane, "Study of children's hugging for interactive robot design," in *2016 25th IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN)*, Aug 2016, pp. 557–561.
- [7] A. Alspach, J. Kim, and K. Yamane, "Design and fabrication of a soft robotic hand and arm system," in *2018 IEEE International Conference on Soft Robotics (RoboSoft)*, April 2018, pp. 369–375.

- [8] R. Niiyama, X. Sun, C. Sung, B. An, D. Rus, and S. Kim, "Pouch motors: Printable soft actuators integrated with computational design," *Soft Robotics*, vol. 2, no. 2, pp. 59–70, 2015. [Online]. Available: <https://doi.org/10.1089/soro.2014.0023>
- [9] R. Niiyama, C. Rognon, and Y. Kuniyoshi, "Printable pneumatic artificial muscles for anatomy-based humanoid robots," in *2015 IEEE-RAS 15th International Conference on Humanoid Robots (Humanoids)*, Nov 2015, pp. 401–406.
- [10] R. Niiyama, X. Sun, L. Yao, H. Ishii, D. Rus, and S. Kim, "Sticky actuator: Free-form planar actuators for animated objects," in *Proceedings of the Ninth International Conference on Tangible, Embedded, and Embodied Interaction*. ACM, 2015, pp. 77–84.
- [11] J. Ou, M. Skouras, N. Vlavianos, F. Heibeck, C.-Y. Cheng, J. Peters, and H. Ishii, "aeromorph-heat-sealing inflatable shape-change materials for interaction design," in *Proceedings of the 29th Annual Symposium on User Interface Software and Technology*. ACM, 2016, pp. 121–132.
- [12] Velmex inc - positioning systems for science and industry. [Online]. Available: <http://www.velmex.com>
- [13] Openbuilds part store. [Online]. Available: <http://openbuildspartstore.com>
- [14] Misumi usa: Industrial configurable components supply. [Online]. Available: <https://us.misumi-ec.com/>
- [15] Convert image pixels to xy-coordinates. [Online]. Available: <https://www.mathworks.com/matlabcentral/fileexchange/54799-convert-image-pixels-to-xy-coordinates>
- [16] Polyurethane products — brake die film urethanes for metal forming and general purpose. [Online]. Available: <http://www.polyprod.com/brake-die-film.html>
- [17] Thermoplastic polyurethane (tpu) typical properties generic tpu-polyether. [Online]. Available: <https://plastics.ulprospector.com/generics/54/c/t/thermoplastic-polyurethane-tpu-properties-processing/sp/11>
- [18] [Online]. Available: <https://www.mcmaster.com>
- [19] A. Delazio, K. Nakagaki, R. Klatzky, S. Hudson, J. Lehman, and A. Sample, "Force jacket: Pneumatically-actuated jacket for embodied haptic experiences," in *Proceedings of the 36th Annual ACM Conference on Human Factors in Computing Systems*, 2018.